

CDRPE: A Combined Deep Learning and Self-Attention Enhanced Reinforcement Learning Framework for Automated Compact Model Parameter Extraction

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As semiconductor technology node advances, the number of parameters in the modern device compact model increases drastically. Manual extraction of these model parameters becomes not only tedious but also impossible, and the automatic method is strongly desired. Traditional black-box optimization suffers from poor scalability due to the curse of dimensionality, while deep learning-based methods typically require large amounts of training data. To address these challenges, we propose CDRPE: a combined deep learning and self-attention enhanced reinforcement learning framework for automatically extracting a large set of DCM parameters across multiple electrical characteristics. The framework leverages a pre-trained multilayer perceptron to initialize core parameters, incorporates device physics knowledge to guide the search, and employs a self-attention-enhanced RL agent for efficient exploration in high-dimensional parameter spaces. Experimental results on BSIM4, BSIMSOI, and BSIMCMG demonstrate that CDRPE can automatically extract 100 parameters with root-mean-square error below 5% relative to TCAD and silicon data. Compared with existing methods, the proposed framework achieves a 7.7x speed up. Moreover, the generated models show good convergence in both digital and analog circuit simulations, exhibiting the potential of this framework for future practical applications.

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1 Introduction

Device compact model (DCM) is a key bridge between semiconductor device technology and circuit design. The turnaround time and accuracy of circuit design simulations are strongly dependent on both the creation cycle and the accuracy of the DCM [1]. In the standard industry DCM modeling practice, DCM parameters are extracted to fit the **technology computer-aided design (TCAD)** or experimental measurement data. As CMOS technology scales into the nanometer regime, non-ideal effects such as **short-channel effects (SCE)** and parasitic effects become increasingly significant [2]. These effects lead to more complex DCM formulations and rapid growth of the number of required fitting parameters in modern DCMs. For example, the **Berkeley Short-Channel IGFET Model 4 (BSIM4)** contains more than 300 parameters, the **BSIM Silicon-On-Insulator (BSIMSOI)** model contains over 400 parameters [3], and the **BSIM Common Multi-Gate (BSIMCMG)** model for advanced FinFET devices includes more than 1000 parameters [4]. Extracting and optimizing such a large number of parameters manually is not only tedious and time-consuming but also increasingly impractical. Therefore, automated methods for DCM parameter extraction are essential for achieving efficient and accurate DCM development.

To tackle this problem, different artificial intelligent techniques including black-box optimization, **deep learning (DL)** and **reinforcement learning (RL)** have been investigated for automated DCM parameter extraction. Black-box optimization algorithms such as **Genetic Algorithm (GA)** [5–10], **Differential Evolution (DE)** [11], **Bayesian Optimization (BO)** [12] and **Particle Swarm Optimization (PSO)** [13, 14] are researched the earliest. These methods have demonstrated strong performance when extracting a limited number of parameters. However, they suffer from parameter search space dimensionality explosion (i.e., curse of dimensionality). As the dimensionality of the parameter space increases, the search space expands exponentially, making efficient exploration and convergence to optimal solutions increasingly difficult. In practice, these algorithms have been applied to extract between 11 [7] and 35 [15] parameters, with some studies extending to approximately 50 parameters [8]. Scaling these methods to the hundreds of parameters remains a substantial challenge. DL-based methods are also developed for DCM parameter extraction [16–20]. In these methods, Monte Carlo simulations are commonly used to generate training datasets covering a wide range of parameter variations. For example, a neural network trained on simulated data is used to achieve rapid extraction of 17 BSIM-CMG parameters [17]. However, the accuracy of DL-based methods relies on a manually extracted baseline model and requires extensive I - V and C - V data for training, resulting in significant measurement overhead that limits practical applicability. More recently, RL algorithms, such as Q-learning, have been investigated for DCM parameter extraction task [21]. However, Q-learning performs better in discrete action spaces, while the DCM parameter search space is inherently continuous. And the number of parameters successfully extracted by Q-learning remains limited to fewer than 20. Notably, a recent study employing the **Asynchronous Advantage Actor-Critic (A3C)** algorithm extracted 51

model parameters, highlighting the potential of RL-based methods in handling high-dimensional DCM parameter extraction tasks [22].

In this work, a **Combined DL and RL framework for DCM parameter extraction (CDRPE)** is proposed to overcome the limitations of existing methods, enable efficient and robust DCM parameter extraction in a continuous action space. Unlike prior approaches that directly extract parameters using black-box optimization or deep learning techniques, the proposed framework adopts a hybrid strategy that combines a **multi-layer perceptron (MLP)** for initial prediction of core parameters and further a **self-attention enhanced RL (SAE-RL)** for efficient fine-tuning of large-scale parameter sets. By leveraging multiple electrical characteristics, CDRPE enables the parameter extraction of various secondary device-physics effects, such as short-channel effects, **channel length modulation (CLE)**, **drain-induced barrier lowering (DIBL)**, quantum machine effect, and so on, thereby facilitating the creation of highly accurate and predictive compact models. The key contributions of this work are summarized as follows:

- (1) We decompose the DCM parameter extraction into initialization and fine-tuning stages, and propose a combined DL and RL framework for automated parameter extraction, enabling the extraction of a larger number of DCM parameters (~100) automatically. This hybrid strategy effectively reduces the search space, enhances sample efficiency and convergence speed toward the global optimum.
- (2) A novel multi-branch self-attention module is incorporated into the RL agent. Each branch focuses on a specific type of electrical characteristic, adaptively weighting features that are most relevant for DCM parameter extraction. This enables effective extraction of parameters for multiple types of electrical characteristics with different physical properties and orders of magnitude, accelerates convergence in high-dimensional optimization and enhances generalization across different DCMs.
- (3) We incorporate device physics knowledge into the state and reward functions of RL agent, ensuring the learning process to occur in the physically meaningful regions of the parameter search space. This improves convergence stability and ensures that the extracted parameters maintain physical interpretability.
- (4) Experiments on industry-standard DCMs (BSIM4, BSIMCMG, and BSIMSOI) demonstrate that the proposed framework is able to automatically extract up to 100 parameters with a **root-mean-square error (RMSE)** of less than 5%. The framework also achieves the fastest convergence speed compared to existing methods. Finally, the created models show great convergency in circuit simulations.

The remainder of this article is organized as follows: Section 2 presents the preliminaries on device compact model, deep learning, reinforcement learning, and attention-based network. Section 3 provides a detailed description of the proposed framework. Section 4 evaluates the performance of the framework through three compact model parameter extraction case studies. Finally, Section 5 concludes the article.

2 Preliminaries

Compact Model Parameters Extraction. As the key component in modern **integrated circuits (ICs)**, DCM consists of formulas and corresponding parameters. It captures the electrical behavior of devices and is further used for SPICE circuit simulation. Traditionally, the optimal parameters of DCM are extracted by sequentially fitting the measured electrical characteristics, such as current and capacitance characteristic of transistor devices under different bias conditions.

Electrical Characteristics. The electrical characteristics of transistor devices determine their performance in ICs. A typical transistor device is composed of four terminals: gate, source, drain, and substrate. The gate and drain biases are denoted as V_g and V_d , respectively. The current flowing from the drain to the source is denoted as I_d , while the gate capacitance is represented by C_{gg} . Noted that both I_d and its derivatives, such as transconductance $g_m = \partial I_d / \partial V_g$ and output conductance $g_d = \partial I_d / \partial V_d$, are critically important for circuit simulations.

Deep Learning. DL is a specialized subset of **Machine Learning (ML)** that uses deep neural networks to model complex patterns in large datasets. The learning process iteratively adjusts the network's internal parameters to minimize a loss function via the backpropagation algorithm and gradient-based optimization. This approach has achieved breakthrough performance in various domains, notably in sequence modeling [23].

Reinforcement Learning. RL is a machine learning paradigm where an agent learns to make optimal decisions through trial-and-error interactions with an environment. The goal is to discover a behavioral policy that maximizes a cumulative reward signal over time. This interaction is formally modeled as an MDP, defined by a tuple (S, A, P, R, γ) , where S, A, P, R, γ is the state, action, state transition probability function, reward function and discount factor, respectively. At each timestep, the agent in state $s \in S$ takes an action $a \in A$ according to its policy $\pi(a|s)$. It then receives a reward $r = R(s, a)$ and transitions to a new state s' based on the probability $P(s'|s, a)$. The agent's objective is to learn an optimal policy π^* that maximizes the expected discounted return. This framework has demonstrated remarkable success in complex domains, including robotic control [24] and mastering Atari games [25].

Attention-Based Network. The Attention-based network enables a model to dynamically weigh the importance of different parts of an input when producing an output, rather than relying on a fixed-length representation [26]. The network typically operates on three vectors derived from the input: the **Query (Q)**, the **Key (K)**, and the **Value (V)**. The output is computed as a weighted sum of the Values, where the weights are determined by the scaled dot-product of Queries and Keys:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V. \quad (1)$$

A key variant is self-attention, where Q, K and V are all projected from the same input sequence. This allows the model to capture the internal dependencies and structural information within the sequence itself. In this framework, we employ self-attention to process the input state, enabling the actor and critic to focus on the most relevant features for decision-making.

3 Methodology

3.1 Problem Formulation

The DCM parameter extraction problem constitutes a fundamental challenge in semiconductor device modeling, where the objective is to determine optimal parameter sets that maximize the agreement between model predictions and target data. This problem can be formulated a finite constraint optimization problem as follows:

$$\begin{aligned} & \underset{x \in \mathbb{R}^n}{\operatorname{argmin}} f(x) \\ & \text{s.t. } x_L^{(i)} \leq x^{(i)} \leq x_U^{(i)}, \quad i = 1, 2, \dots, n \end{aligned} \quad (2)$$

where x is an n -dimensional DCM parameter vector, $x^{(i)}$ is the i th component of x , $\mathbb{R}^n$ represents the n -dimensional parameter search space, x_L and x_U are the upper and lower bounds of the parameters, respectively. The objective function $f(x)$ evaluates the performance of x with respect to

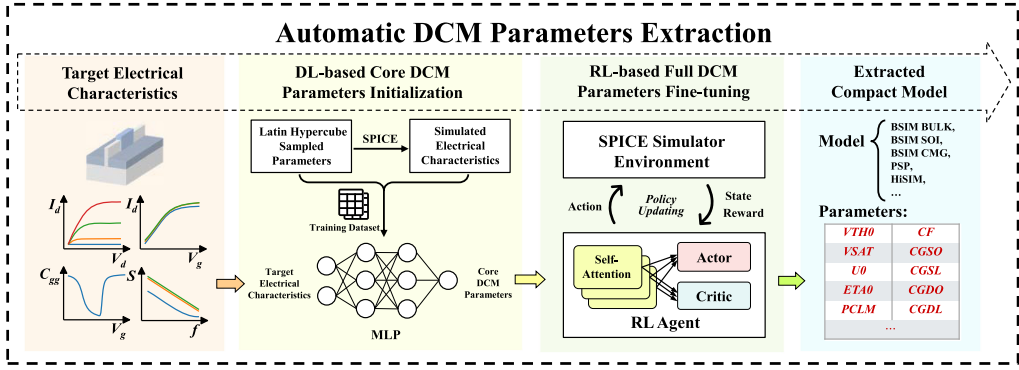


Fig. 1. Schematic diagram of automatic DCM parameters extraction based on CDRPE.

multiple target electrical characteristics, and is defined as a weighted sum of **root-mean-square errors (RMSEs)** across different characteristics:

$$f(x) = \sum_{i=1}^{N_{char}} w_i \cdot \left(\frac{1}{N_i} \sum_{j=1}^{N_i} \left(1 - \frac{y_{i,j}^{model}(x)}{y_{i,j}^{target} + \varepsilon} \right)^2 \right)^{\frac{1}{2}}, \quad (3)$$

where N_{char} denotes the number of distinct electrical characteristics, w_i is the weight assigned to the i th characteristic, N_i is the number of data points associated with that characteristic, and ε is a predefined constant. Here, $y_{i,j}^{target}$ represents the reference measurement of the j th data point in the i th characteristic, while $y_{i,j}^{model}(x)$ denotes the corresponding model-generated value under the parameter set x .

3.2 Framework Overview

An overview of the DCM parameter extraction framework is shown in Figure 1. Traditional parameter extraction methods typically require sequential fitting of various characteristics, such as I_d-V_d , I_d-V_g and $C_{gg}-V_g$ under different bias conditions. This process often demands substantial manual effort and iterative optimization. Particularly, engineers typically begin by adjusting the parameters that significantly influence electrical characteristics, such as threshold voltage (V_{TH0}), mobility (U_0), and so on, referred to as core DCM parameters. The supplementary parameters that capture various secondary effects are subsequently adjusted. By iteratively optimizing both core and supplementary parameters, the DCM achieves a more precise fitting across different operating conditions.

To mimic this process, CDRPE is designed to handle the DCM parameter extraction process. First, an MLP is trained on a dataset generated by HSPICE simulations of the DCM, where the parameter space is explored through **Latin Hypercube Sampling (LHS)**. The network is designed to capture the nonlinear relationship between the electrical characteristics and the corresponding core DCM parameters. Once pre-trained, the MLP takes the target electrical characteristics as input and outputs an initial set of core DCM parameters. These parameters are then employed in a SAE-RL framework in two aspects: (1) it is applied to the DCM to generate the initial electrical characteristics, which serve as the starting state of the RL agent; (2) it serves as the starting point of the parameter optimization trajectory. Subsequently RL agent performs further fine-tuning of the complete set of DCM parameters to achieve a closer match to the target characteristics. Unlike black-box optimization methods, which operate directly in the full parameter space and suffer from exponential growth in search complexity. The SAE-RL formulates the DCM parameter

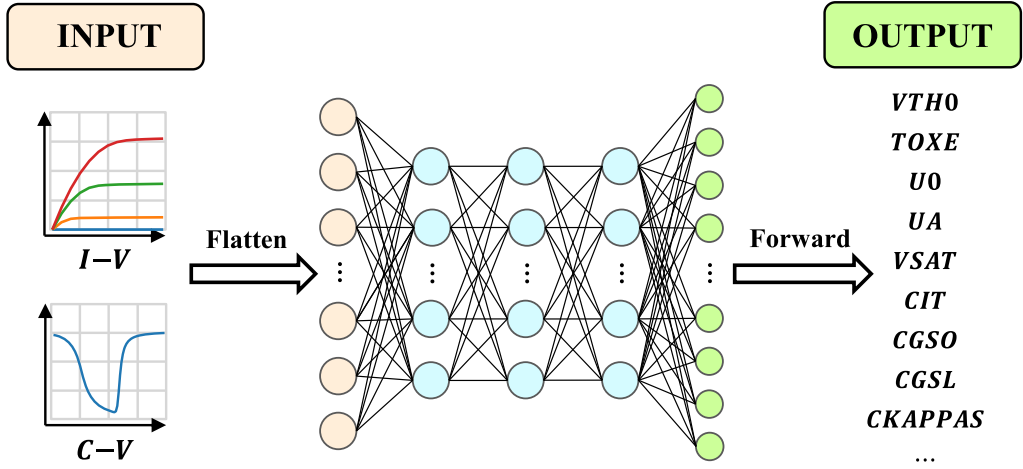


Fig. 2. The structure of the pre-trained MLP. The MLP takes current and capacitance characteristics as input and predicts core parameters, which are employed to initialize the action space and generate the initial state for subsequent RL fine-tuning.

fine-tune task as an MDP (S, A, P, R, γ) , where a state $s \in S$ comprises, multiple electrical characteristics generated from the current DCM parameter vector, and an action $a \in A$ corresponds to the change of DCM parameters. The agent receives electrical characteristics as input, which are processed by a multi-branch self-attention module to extract the most informative feature representations for DCM parameter extraction. These representations are then served as input to both the actor and critic networks, enabling accurate parameter optimization through iterative interaction with a SPICE simulation environment. In addition, device physics knowledge is integrated into the state representation and reward function, constraining the learning process to the physically meaningful regions of the parameter search space.

3.3 DL-based Initialization of Core DCM Parameters

To estimate the initialization of core DCM parameters, we formulate this task as a supervised learning problem, where the goal is to learn a mapping from electrical characteristics to core DCM parameters.

The selection of core DCM parameters depends on its influence on the electrical characteristics and specific DCMs. For example, in the case of BSIM4, the core DCM parameters are related to threshold voltage, equivalent oxide thickness, mobility, velocity saturation, DIBL, SCE, extension resistance and overlap capacitance, as follows: V_{TH0} , $TOXE$, U_0 , U_A , CIT , $VSAT$, $ETA0$, $PVAG$, $PCLM$, $DVT0$, $RDSW$, $CGSO$, $CGSL$, $CKAPPAS$, $CGDO$, $CGDL$, and $CKAPPAD$.

These parameters facilitate an approximate fitting of the device's electrical behavior, with each parameter predominantly affecting specific bias regions. For examples, U_0 , U_A , CIT , and $RDSW$ mainly affect the linear region of I_d-V_g , $ETA0$, $PVAG$, and $DVT0$ mainly affect the saturation region of I_d-V_g . $VSAT$ and $PCLM$ primarily influence the saturation region of I_d-V_d . The overlap capacitance parameters affect $C_{gg}-V_g$ under different V_d . To balance modeling accuracy with computational efficiency, only 15–20 core DCM parameters are pre-trained.

The MLP is training on a synthetic dataset generated via LHS. This stratified random sampling technique is selected over conventional random sampling to ensure a more uniform and comprehensive coverage of the high-dimensional parameter space [27]. Parameters are sampled

Table 1. The Range of Core DCM Parameters in BSIM4 Model for Latin Hypercube Sampling

Core DCM Parameters	Range	Description
$VTH0(V)$	(0.2, 0.5)	Threshold voltage
$TOXE(nm)$	(2, 5)	Equivalent oxide thickness
$U0(m^2/(V \cdot s))$	(50, 600)	Low field mobility
$UA(m/V)$	$(-5 \times 10^{-9}, 5 \times 10^{-9})$	Phonon roughness scattering parameter
$VSAT(m/s)$	$(8 \times 10^4, 1.2 \times 10^5)$	Saturation velocity
$ETA0$	(0, 2)	DIBL coefficient in subthreshold region
$PVAG$	(0, 1.5)	Gate-bias dependence on early voltage
$PCLM$	(0.1, 1.5)	Channel length modulation parameter
$CIT(F/m)$	$(0, 3 \times 10^{-3})$	Interface trap capacitance
$DVT0$	(1, 3)	First coefficient of short-channel Effect on threshold voltage
$RDSW(\Omega \cdot \mu m^{W_r})$	(0, 200)	Zero bias LDD resistance per unit width
$CGSO(F/m)$	$(0, 5 \times 10^{-11})$	Non LDD region source-gate overlap capacitance per unit channel width
$CGSL(F/m)$	$(0, 5 \times 10^{-11})$	Overlap capacitance between gate and lightly-doped source
$CKAPPAS$	(0.3, 0.9)	Coefficient of overlap capacitance
$CGDO(F/m)$	$(0, 5 \times 10^{-11})$	Non LDD region drain-gate overlap capacitance per unit channel width
$CGDL(F/m)$	$(0, 5 \times 10^{-11})$	Overlap capacitance between gate and lightly-doped source region
$CKAPPAD$	(0.3, 0.9)	Coefficient of overlap capacitance

from predefined ranges (Table 1), and corresponding I_d-V_g , I_d-V_d , and $C_{gg}-V_g$ characteristics are obtained through SPICE simulations under multiple device geometries (different W/L ratio). This process guarantees that the dataset captures a wide spectrum of realistic device behaviors, thereby enhancing the generalization ability of the model. This dataset comprises 50,000 samples of electrical characteristics, i.e., 50,000 complete sets of I-V and C-V curves where each set is generated from one sampled parameter vector. Since geometry variations are already incorporated during pre-training, the trained MLP is directly reused for parameter extraction across different device geometries within the same process node, without any further retraining.

While I-V and C-V characteristics exhibit significantly differences in numerical values and distributions, a unified MLP is employed for the parameter extraction task. This design ensures physical consistency of the predicted parameters, as many parameters jointly influence both I-V and C-V behaviors. To mitigate scale disparities, all input characteristics are standardized via Z-score normalization and flattened into a single input vector.

The MLP architecture consists of one input layer, three hidden layers with 256 neurons each, and an output layer, with tanh activations applied between layers. The MLP is trained using the **Backpropagation (BP)** algorithm, with the **mean square error (MSE)** between predicted and target parameters as the loss function:

$$\mathcal{L}_{MSE} = \frac{1}{N} \sum_{i=1}^N \|y_i - \hat{y}_i\|_2^2. \quad (4)$$

where N denotes the batch size, y_i and $\hat{y}_i$ represent the target and predicted vector of the i th sample, respectively. To reduce non-uniqueness, the MLP simultaneously processes multiple current and capacitance data points under different bias conditions [17], enabling the model to capture parameter sensitivities across bias regions. The details is presented in Algorithm 1.

ALGORITHM 1: DL-based MLP Training Procedure**Input:** IV, CV .**Output:** Core DCM parameters.

- 1: Randomly initialize the parameters of $P_{IV}(z_{IV}|\theta_{IV}^{P_{IV}})$ and $P_{CV}(z_{CV}|\theta_{CV}^{P_{CV}})$;
- 2: **repeat**
- 3: **for** $i = 1$ **to** N **do**
- 4: Calculate the output of y'_{IV} and y'_{CV} ;
- 5: Update pretraining network by minimizing the loss function loss_{IV} and loss_{CV} ;
- 6: **end for**
- 7: **until stop condition**

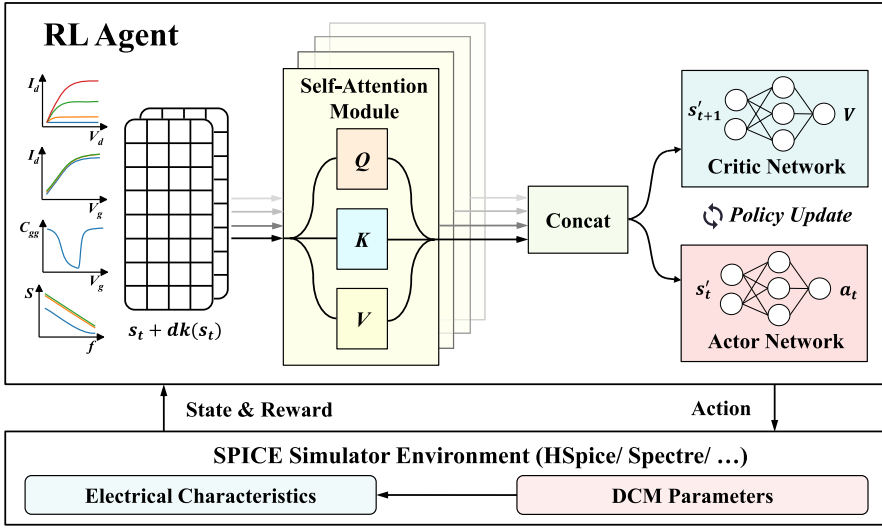


Fig. 3. Detailed diagram of automatic parameter extraction based on SAE-RL.

The trained MLP predicts an initial set of core DCM parameters given target current and capacitance characteristics. These parameters, together with the remaining DCM parameters, serve as the initialization point for fine-tuning using a continuous-action SAE-RL framework. Specifically, they are employed to generate the corresponding electrical characteristics through SPICE simulation, and the simulated characteristics, together with the initial parameters, serve as the initial state $s_{initial}$ and action $a_{initial}$ for the RL framework. Starting from this informed initialization, the RL agent performs continuous fine-tuning to iteratively refine the parameters, thereby improving training efficiency, accelerating convergence, and achieving a closer match between model outputs and target characteristics.

3.4 Self-attention Enhanced RL Formulation

While the pre-trained MLP provides initial estimates of the core DCM parameters, achieving precise matching with the target electrical characteristics requires further fine-tuning of the majority of DCM parameters. This fine-tuning is performed by an RL agent, and the detailed procedure is illustrated in Figure 3. The parameter extraction task is formulated as an MDP, which proceeds

according to the following sequence:

$$s_{initial} \xrightarrow{a_0} (s_1, r_1) \xrightarrow{a_1} \dots \xrightarrow{a_{t-1}} (s_t, r_t), \quad (5)$$

where s_i , a_i and r_i represent the state, action and reward in step t . To handle the heterogeneous electrical characteristics, a multi-branch self-attention module is embedded within the RL agent. Each branch extracts features from a specific type of electrical characteristic, projecting them into Q , K , and V vectors. Attention weights are then computed as the scaled dot-product between Q and K , producing attention vectors that emphasize the most informative features for DCM parameter extraction. These feature representations are subsequently concatenated to s'_t , which serves as input of actor and critic networks. The agent continuously updates its policy based on the states and the rewards received. The Actor network adjusts the policy according to the value function estimated by the Critic network. And the Critic network refines the value function estimates. This iterative process allows for the convergence of the agent's behavior toward an optimal policy over time. The set-up of four key elements of the MDP are defined as follows:

State Space. State represents the device electrical characteristics, including output, transfer, capacitance and noise characteristics. These characteristics are obtained from the SPICE simulation environment with the DCM parameters predicted by the Actor network. In the framework, the state is constructed as a multi-dimensional tensor, where each dimension corresponds to a specific type of electrical characteristic:

$$State = [I_{out}, I_{tran}, C_{gg}, Noise]^T. \quad (6)$$

Typically, these device characteristics can exhibit linear or nonlinear behavior, across multiple orders of magnitude. This can hinder the training process, leading to convergence issues or training failure. To address this issue, a Box-Cox transformation is applied to the data, which normalizes the distribution and compresses extreme values, thereby accelerating model convergence.

Action Space. In this framework, the action is defined as the incremental change of each DCM parameter rather than its absolute value. This incremental design provides smooth updates and mitigates the risk of divergence, especially under gradient-based policy updates. The action vector has the same dimension as the DCM parameter vector:

$$Action = [\Delta U0, \Delta VTH0, \Delta VSAT, \dots]^T. \quad (7)$$

In MDP, the DCM parameters input to the SPICE simulation environment at time step t are updated by accumulating all previous increments:

$$\begin{aligned} x_t &= a_{initial} + \sum_{i=0}^t a_i \\ &= a_{initial} + \sum_{i=0}^t [\Delta VTH0^{(i)}, \Delta U0^{(i)}, \dots]^T \\ &= [VTH0, U0, \dots]^T. \end{aligned} \quad (8)$$

Specially, $a_{initial}$ is the initial parameters derived from the pre-trained MLP and technical manual. The action range is further refined by a dynamic shrinking strategy to improve stability. Specifically, the action bounds for the i th parameter at epoch n_{epoch} are defined as

$$[a_L^{(i)}, a_U^{(i)}] = (1 - n_{epoch} \gamma) [x_L^{(i)}, x_U^{(i)}], \quad (9)$$

where γ is a decay factor. Note all of these action values are normalized between -1 and 1 to facilitate smooth network training and improve convergence speed. for linear spacing, the parameter range (x_L, x_U) is mapped to $(-1, 1)$ as

$$a_{norm} = 2 \cdot \frac{a - a_L}{a_U - a_L} - 1, \quad (10)$$

which results in more uniform sampling across different orders of magnitude, improving exploration in small-value regions. In the case of parameter extraction tasks, the range of parameters typically isn't large, and the linear mapping leads more stable training.

Compared to the discrete action space RL algorithms, continuous action spaces allow for finer granularity in parameter adjustments, enabling more precise control over the optimization process. This enhances convergence rates and improves the performance of parameter extraction tasks.

Reward. Reward is the optimization objective that guides the agent toward improved policies. It quantifies the quality of an action taken in a given state and is defined to reflect the alignment between the simulated model output and the target data. A larger reward indicates a closer match between the model output and the target data, which also indicates the better the RL agent learns from its iterations. The reward function is carefully designed to balance three key objectives: simulation stability, fitting accuracy, and physical plausibility. The overall reward for a given DCM parameter vector x is expressed as

$$R(x) = R_s(x) + R_f(x) + R_r(x). \quad (11)$$

$R_s(x)$ represents the simulation constraint, penalizes simulation failures in the SPICE environment, ensuring that the extracted parameter set remains within a valid operating region:

$$R_s(x) = \begin{cases} -1, & \text{if SPICE simulation fails} \\ 0, & \text{otherwise.} \end{cases} \quad (12)$$

$R_f(x)$ denotes the fitting accuracy reward. It is defined as the negative weighted sum of RMSEs across different characteristics:

$$R_f(x) = -f(x) = - \sum_{i=1}^{N_{char}} w_i \cdot \left(\frac{1}{N_i} \sum_{j=1}^{N_i} \left(1 - \frac{y_{i,j}^{model}(x)}{y_{i,j}^{target} + \varepsilon} \right)^2 \right)^{\frac{1}{2}}. \quad (13)$$

To prevent convergence toward non-physical or extreme parameter values, a soft regularization term $R_r(x)$ is applied:

$$R_r(x) = -\lambda \cdot \|x - x_{mid}\|^2, \quad (14)$$

where x_{mid} is a reference vector derived from technical documentation or expert knowledge, and λ is a tunable coefficient that controls the strength of regularization. A large λ enforces conformity to the reference values, stabilizing exploration but potentially leading to conservative updates; conversely, a small λ weakens this physical constraint, allowing for broader exploration but increasing the risk of unphysical solutions.

Actor and Critic. The RL agent follows an Actor-Critic framework, as illustrated in Figure 3. The Actor network receives the current state and outputs an action to update the DCM parameters, while the Critic network evaluates the expected return of the current state. Both networks are trained jointly: the Critic guides the Actor by estimating the value function, and the Actor updates the policy based on the estimated advantage. The iterative optimization continues a predefined target RMSE is achieved.

The current, capacitance and noise characteristics exhibit different physical properties and orders of magnitude, making it difficult to process multiple characteristics with a single MLP, resulting in numerical instability and poor convergence. Moreover, the strong coupling among DCM

parameters prevents the multi-agent approach where each agent handles a characteristic independently. To address these challenges, the Actor and Critic networks share a multi-branch self-attention module. Each branch processes a specific type of electrical characteristic, avoiding different orders of magnitude being processed by the same set of parameters. Attention weights are then computed as the scaled dot-product between Q and K as shown in Equation (15). The scaling factor d serves as a safeguard against gradient explosion, ensuring that the dot-product scores remain within a stable numerical range. The subsequent *softmax* operation converts these scores into a probability distribution, a set of weights between 0 and 1 that sum to 1, thereby capturing the relative importance of different data point. The resulting feature representations from all branches are then concatenated and input to both Actor and Critic networks. This module highlights the most informative features within each characteristic, effectively learning domain-specific representations while mitigating numerical instability caused by vastly different orders of magnitude.

$$h^{(i)} = \text{Softmax} \left(\frac{QK}{\sqrt{d}} \right) V, \quad (15)$$

$$H = \text{Concat} \left(h^{(1)}, h^{(2)}, \dots \right), \quad (16)$$

where $h^{(i)}$ denotes the feature representation from the i th characteristic domain, and H is the concatenated representation fed into both networks. This design leverages the self-attention mechanism to assign adaptive weights to different electrical characteristics, extracting features most relevant to DCM parameters. The resulting representations enhance the efficiency and accuracy of subsequent RL-based parameter optimization by providing the Actor and Critic networks with high-quality and informative inputs.

Following feature extraction, both Actor and Critic networks are implemented as MLPs comprising an input layer, two hidden layers (256 and 128 neurons, respectively), and an output layer, with tanh activation functions between layers [28]. For continuous action spaces, the Actor network receives the current state s_t as input and outputs a probability distribution P_t over possible actions. Next, the action a_t is obtained by sampling from this distribution P_t . The action a_t is then sampled from P_t , ensuring stochastic behavior and encouraging exploration. The next state s_{t+1} and reward r_t are subsequently observed from the environment.

The Critic network takes the next state s_{t+1} as input and outputs the estimated value function, defined as $V^\pi(s|\theta^V)$, where π is the current policy and θ^V denotes the parameters of critic network. According to the Bellman equation, the relationship between the value of the current state and next state can be established

$$V^\pi(s_t) = \mathbb{E}_{a_t \sim \pi} [r_t + \gamma V^\pi(s_{t+1})], \quad (17)$$

where $\gamma \in [0, 1]$ is discount factor. The value function quantifies the expected return of the current state and is used to compute the Critic loss, which is defined as the **temporal-difference (TD)** error:

$$L^V(\theta^V) = \left(r_t + \gamma V^\pi(s_{t+1}|\theta^V) - V^\pi(s_t|\theta^V) \right)^2. \quad (18)$$

The Actor network updates its policy based on the advantage function [29]:

$$A_t = r_t + \gamma V^\pi(s_{t+1}) - V^\pi(s_t), \quad (19)$$

which measures the relative improvement of the chosen action compared to the expected value. This joint training mechanism enables the Actor to refine the policy iteratively while the Critic provides stable value estimates, facilitating efficient convergence in high-dimensional continuous parameter spaces.

3.5 Device Physics Knowledge Integration

Incorporating the domain knowledge into RL agent enhance the agent's ability to learn optimal policy and reach the desired objective [30]. In the context of DCM parameter extraction, certain parameters have a stronger effect on the derivatives of electrical characteristics. Here we propose integrating device knowledge g_m and g_{ds} into the RL algorithms. Specifically, the transconductance g_m and output conductance g_{ds} provide higher-order information that is highly sensitive to variations in critical device parameters.

The proposed framework integrates g_m and g_{ds} into RL agent. These features are denoted as device knowledge $dk(s)$, is incorporated to the state and reward as additional input features for policy learning. In particular, the Critic network utilizes $dk(s)$ in the value function to enhance the accuracy of value estimation:

$$Q_i \leftarrow r_i + \gamma V(s_{i+1}, dk(s_{i+1}) | \theta^V), \quad (20)$$

where θ^V denotes the parameters of the value network. Since the policy update in Actor-Critic methods is derived from the Critic's value estimation, embedding device knowledge directly influences both policy evaluation and improvement. Consequently, the agent can more efficiently identify parameter adjustments that reduce the discrepancy between target and simulated characteristic, guiding the agent toward regions of the parameter space that are physically meaningful and obtain high-quality solutions.

3.6 Algorithm Execution Flow

The detailed flow of DCM parameter fine-tuned is summarized in Algorithm 2. At the beginning, the Actor and Critic network are initialized with randomized values, and a replay buffer D with a capacity of N transitions is created to store transition tuples. The RL agent is initialized with state $s_{initial}$ and action $a_{initial}$.

The optimization process proceeds over T epoch. In each epoch, the transition (s_t, a_t, r_t, s_{t+1}) is stored in the replay buffer. At time step t , the current state is first processed by a multi-branch self-attention network, provided as input to both the Actor and Critic. The Actor network generates an action $a_t \sim \pi_\theta(a_t | s_t)$ to update the DCM parameters. The environment returns the next state s_{t+1} and a reward r_t , which quantifies the accuracy of the model output relative to the target electrical characteristics. Each transition is stored in the replay buffer.

Once the buffer contains N transitions, mini-batches are drawn to update the networks. The Critic network evaluates the value of each state, while the Actor updates its policy based on this evaluation. The self-attention network is updated jointly through feedback from both networks, ensuring that the extracted features support effective policy learning.

This iterative optimization continues until a predefined convergence criterion is met. By incorporating device physics knowledge and leveraging shared feature extraction within a continuous-action reinforcement learning framework, the framework efficiently refines the DCM parameters. During the optimization, the agent adaptively balances exploration and exploitation, which accelerates convergence and enhances the accuracy and physical consistency of the extracted parameters.

4 Experiments

To demonstrate the effectiveness of the automated DCM parameters extraction framework, multiple experiments are conducted to compare the performance of the proposed framework with existing optimization methods on different industrial standard DCMs. The proposed framework is implemented in Pytorch framework. The industry standard Synopsys HSpice simulator is used for all experiments on Intel Xeon Gold 6248R CPU with a base frequency of 3.0 GHz.

ALGORITHM 2: SAE-RL-based DCM parameter extraction**Input:** IV, CV **Output:** DCM parameters.

```

1: Randomly initialize Actor network  $\pi(s|\theta^\pi)$  and Critic network  $V(s|\theta^V)$ ;
2: Initialize the replay buffer  $D$  batch size  $B$  and maximum epoch  $T$ ;
3: Select initial action  $a_0$  from pre-trained MLP and device manual values;
4: Observe the initial state  $s_0$ ;
5: for  $t=1$  to  $T$  do
6:   for  $i$ -th electrical characteristic in  $s_t$  do
7:      $Q_i \leftarrow \pi_Q(s_t), K_i \leftarrow \pi_K(s_t), V_i \leftarrow \pi_V(s_t)$ ;
8:      $h_i = \text{Softmax}\left(\frac{Q_i K_i}{\sqrt{d}}\right) V$ ;
9:     Aggregate feature representation  $s'_t = \text{Concat}(h^{(1)}, h^{(2)}, \dots)$ ;
10:    Select action  $a_t \leftarrow \pi(s'_t; \theta^\pi)$ ;
11:    Interact with environment to obtain the next state  $s_t$  and reward  $r_t$ ;
12:    Store transition  $(s_t, a_t, r_t, s_{t+1})$  in replay buffer  $D$ ;
13:    if  $\text{size}(D) > B$  then
14:      Sample a batch of  $B$  transitions  $(s_i, a_i, r_i, s_{i+1})$  in  $D$ ;
15:       $y_i \leftarrow r_i + \gamma V(s_{i+1}, dk(s_{i+1})|\theta^V)$ ;
16:       $A_i \leftarrow y_i - V(s_i, dk(s_i)|\theta^V)$ ;
17:      for  $j=1$  to  $T$  do
18:        Update Critic network by minimizing the TD error:
19:        
$$L = \frac{1}{B} \sum \left( y_i - V(s_i, dk(s_i)|\theta^V) \right)^2$$
;
20:        Update Actor network by minimizing the loss function:
21:        
$$L = E^t \left[ \min \left( \frac{\pi(s_i, dk(s_{i+1})|\theta^\pi)}{\pi(s_i, dk(s_{i+1})|\theta_{old}^\pi)} A_i, \text{clip} \left( \frac{\pi(s_i, dk(s_{i+1})|\theta^\pi)}{\pi(s_i, dk(s_{i+1})|\theta_{old}^\pi)}, 1 - \epsilon, 1 + \epsilon \right) A_i \right) \right]$$
;
22:      end for
23:      Empty the replay buffer  $D$ ;
24:    end if
25:  end for

```

CDRPE is compared with GA, DE, BO, and conventional RL techniques, including DDPG and PPO. The hyperparameters of algorithms are determined through grid search, as summarized in Table 3. The Adam optimizer is employed. Each experiment is repeated ten times, and the median performance is chosen for comparison. DCM parameter extraction task based on BSIM4 model is tested for different algorithms in the first experiment. Furthermore, to validate the physics consistency of the extracted parameters, the DCM parameters are extracted from I - V and C - V characteristics of devices with different device width (W) and length (L). Then similar experiments are conducted on the BSIMCMG model for FinFET and the BSIMSOI model for SOI MOSFET. Both TCAD simulation data and measured data are extracted to further demonstrate the robustness of the proposed framework. Finally, circuit-level simulations are performed using the extracted models.

4.1 DCM Parameter Extraction for BSIM4

A 54-parameters extraction task based on the BSIM4 model is performed using TCAD data at $W = 0.8\mu\text{m}$ and $L = 0.2\mu\text{m}$. The simulation data includes output, transfer, capacitance and noise characteristic under various bias conditions. To ensure accurate model fitting, 54 DCM parameters are selected, including 17 core parameters, 37 parameters related to current, capacitance and noise characteristic, as shown in Table 1 and 2.

Table 2. The Range of DCM Parameters in BSIM4 Model

Parameters	Range	Description
<i>WINT</i>	$(0, 1 \times 10^{-8})$	Channel-width offset parameter
<i>LINT</i>	$(0, 5 \times 10^{-9})$	Channel-length offset parameter
<i>UB</i>	$(1 \times 10^{-19}, 1 \times 10^{-17})$	Coefficient of second-order mobility degradation
<i>EU</i>	(1, 2.5)	Exponent for mobility degradation
<i>DVT1</i>	(0.3, 0.7)	Second coefficient of short-channel effect on V_{th}
<i>DSUB</i>	(0.4, 1)	DIBL coefficient exponent in subthreshold region
<i>DROUT</i>	(0.4, 1)	Channel-length dependence of DIBL effect on R_{out}
<i>MINV</i>	(-0.5, 0.5)	V_{gsteff} fitting parameter for moderate inversion condition
<i>PHIN</i>	(0, 0.3)	Non-uniform vertical doping effect on surface potential
<i>CDSC</i>	$(0, 3 \times 10^{-4})$	coupling capacitance between source/drain and channel
<i>CDSCD</i>	$(0, 3 \times 10^{-4})$	Drain-bias sensitivity of CDSC
<i>NFACTOR</i>	(0.5, 1.5)	Subthreshold swing factor
<i>VOFF</i>	(-0.2, 0)	Offset voltage in subthreshold region
<i>PDIBLC1</i>	(0, 0.5)	Parameter for DIBL effect on R_{out}
<i>PDIBLC2</i>	(0, 0.05)	Parameter for DIBL effect on R_{out}
<i>RDSWMIN</i>	(0, 100)	LDD resistance per unit width at high V_{gs} and zero V_{bs}
<i>K1</i>	(0, 0.5)	First-order body bias coefficient
<i>K2</i>	(0, 0.05)	Second-order body bias coefficient
<i>ALPHA0</i>	$(0, 1 \times 10^{-6})$	First parameter of impact ionization current
<i>BETA0</i>	(0, 50)	First V_{ds} dependent parameter of impact ionization current
<i>AGIDL</i>	$(0, 1 \times 10^{-9})$	Pre-exponential coefficient for GIDL
<i>BGIDL</i>	$(1 \times 10^9, 5 \times 10^9)$	Exponential coefficient for GIDL
<i>DLC</i>	$(0, 1 \times 10^{-8})$	Channel-length offset parameter for CV model
<i>DWC</i>	$(0, 5 \times 10^{-9})$	Channel-width offset parameter for CV model
<i>NOFF</i>	(0.5, 3)	CV parameter in $V_{gsteff,CV}$ for weak to strong inversion
<i>VOFFCV</i>	(-0.2, 0)	CV parameter in $V_{gsteff,CV}$ for week to strong inversion
<i>ACDE</i>	(0, 2)	Exponential coefficient for charge thickness
<i>MOIN</i>	(5, 20)	Coefficient for the gate-bias dependent surface potential
<i>NOIA</i>	$(1 \times 10^{39}, 1 \times 10^{42})$	Flicker noise parameter
<i>NOIB</i>	$(1 \times 10^{24}, 1 \times 10^{27})$	Flicker noise parameter
<i>NOIC</i>	$(1 \times 10^7, 1 \times 10^9)$	Flicker noise parameter
<i>EM</i>	$(1 \times 10^6, 1 \times 10^8)$	Saturation field
<i>EF</i>	(0.8, 1.2)	Flicker noise frequency exponent
<i>RNOIA</i>	(0, 1)	Thermal noise parameter
<i>RNOIB</i>	(0, 1)	Thermal noise parameter
<i>TNOIA</i>	(0, 2)	Thermal noise parameter
<i>TNOIB</i>	(0, 5)	Thermal noise parameter

Figure 4(a) illustrates the comparison of the objective function $f(x)$ between traditional optimization method and the proposed method in this work. The optimization task is considered successfully completed once the objective satisfies $f(x) < 0.05$, as indicated by the dash-dot reference line. The training of the pre-trained MLP is considered a one-time cost and is not included in the extraction runtime. During the optimization phase, the inference time of the pre-trained MLP is negligible (< 0.05 s) compared to the time cost of SPICE simulations. The proposed framework CDEPR demonstrates superior performance compared with all other methods. Specially, CDRPE

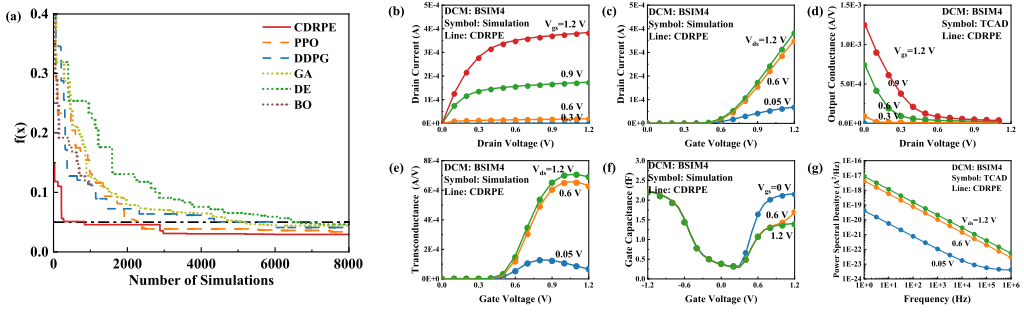


Fig. 4. (a) Performance comparisons for 54-parameter extraction task of BSIM4 model. (b)–(g) are the model (lines) and data (symbols) comparisons of I_d – V_d , I_d – V_g , g_{ds} – V_d , g_m – V_g , C_{gg} – V_g and Power Spectral Density curves.

Table 3. Training Hyperparameter Settings

Hyperparameter	Value
Actor learning rate	3×10^{-4}
Critic learning rate	1×10^{-3}
Discount factor (γ)	0.95
Clip range (ϵ)	0.15
Replay buffer size (D)	5,000
Batch size (B)	128
Regularization coefficient (λ)	0.99

Table 4. Convergence Results of Different Parameter Extraction Methods

Method	Converged ($f(x) < 0.05$)	RMSE	Number of iterations	Runtime (s)
Manual	Yes	4.37%	\	$\sim 5 \times 10^3$
CDRPE	Yes	2.94%	842	75.67
PPO	Yes	3.36%	2295	201.22
DDPG	Yes	4.09%	4354	384.97
GA	Yes	4.63%	5159	452.32
DE	Yes	4.40%	6421	580.56
BO	No	11.3%	1000	$> 10^6$

is 2.7–5.1 times faster than PPO and DDPG, and 6.4–7.7 times faster than black-box optimization techniques such as DE and GA, as shown in Table 4.

BO is widely adopted in optimization tasks and serves as a standard baseline, demonstrating superior convergence speed and accuracy for low-dimensional problems. However, its efficiency significantly decreases as the dimensionality of the parameter space increases. In this experiment, BO is terminated after 1,000 iterations (~ 30 hours) using the same hardware as other algorithms. The runtime is significantly longer than that of the other algorithms, as shown in Table 4, highlighting that the computational cost of BO is prohibitively high for large parameter sets. For the other algorithms, the runtime is dominated by SPICE simulation rather than by the optimization procedure itself. Consequently, the overall computation time can be considered approximately proportional to the number of simulations.

Table 5. Comparison of Automatically and Manually Extracted Parameters of BSIM4 Model

Parameters	Automatically Extracted Results	Manually Extracted Parameters
$V_{TH}(V)$	0.38	0.39
$U_0(m^2/(V \cdot s))$	1.18×10^{-2}	1.32×10^{-2}
$U_A(m/V)$	-2.18×10^{-9}	-2.08×10^{-9}
$VSAT(m/s)$	8.15×10^4	8.64×10^4
ETA_0	8.52×10^{-2}	8.97×10^{-2}
$PCLM$	1.29	1.15
$PVAG$	1.27	1.2
$RDSW(\Omega\mu m^{W_r})$	159	100
DVT_0	1.19	1.5
$TOXE(nm)$	0.25	0.24
K_1	0.20	0.27
K_2	1×10^{-3}	1.67×10^{-3}
$DROUT$	0.55	0.56
...		

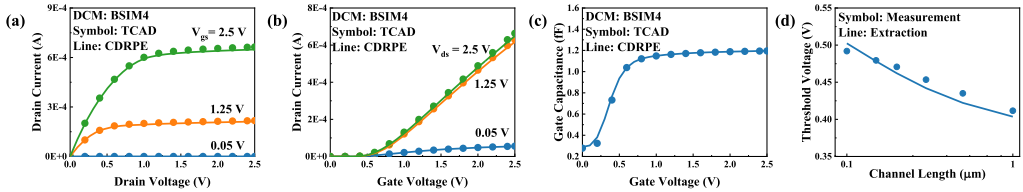


Fig. 5. (a)–(c) are the model (lines) and data (symbols) comparisons of I_d-V_d , I_d-V_g and $C_{gg}-V_g$ predicted results for $W = 1\mu m$, $L = 0.25\mu m$. (d) is the measured and extracted threshold voltages vs. channel length.

Figure 4(b)–(g) show the model outputs generated by CDRPE in comparison with the simulation data of I_d-V_d , I_d-V_g , $\lg(I_d)-V_g$, $g_{ds}-V_d$, g_m-V_g , $C_{gg}-V_g$ and Power Spectral Density. The RMSE of current, capacitance and noise characteristics are 1.76%, 4.08%, and 0.34%, respectively. The extracted parameters are compared with manually extracted parameters shown in Table 5. The majority of automatically extracted parameters differ from manually extracted parameters by less than 10%, indicating that the proposed framework not only ensures fitting accuracy but also preserves the physical interpretability of the compact model.

Optimal parameters can enhance the extendibility of the DCMs, allowing the models to accurately predict electrical characteristics across devices with different W/L . This is a significant advantage for the physics-based DCM in analog circuit design practice. To evaluate this capability, four sets of electrical characteristics are generated based on TCAD with different W/L of 0.8/0.2, 0.8/0.3, 1.2/0.2, 1.2/0.3 ($\mu m/\mu m$), respectively. In addition to the original 54 DCM parameters, geometric scaling parameters, such as $LVTH_0$, $WVTH_0$, $PVTH_0$, LU_0 , WU_0 , and PU_0 are incorporated, resulting in a total of 79 DCM parameters. After parameter extraction using the proposed CDRPE framework, the $f(x)$ reached 0.0439 after 2,146 iterations. The created model is further validated by predicting electrical characteristics with unseen W/L , as shown in Figure 5(a)–(c), achieving RMSE values below 5% for both $I-V$ and $C-V$ characteristics relative to TCAD simulation results. Moreover, Figure 5(d) compares the extracted threshold voltages using the constant-current method in the linear region across devices with different lengths, demonstrating strong agreement. These results confirm the physical validity and extendibility of the proposed framework.

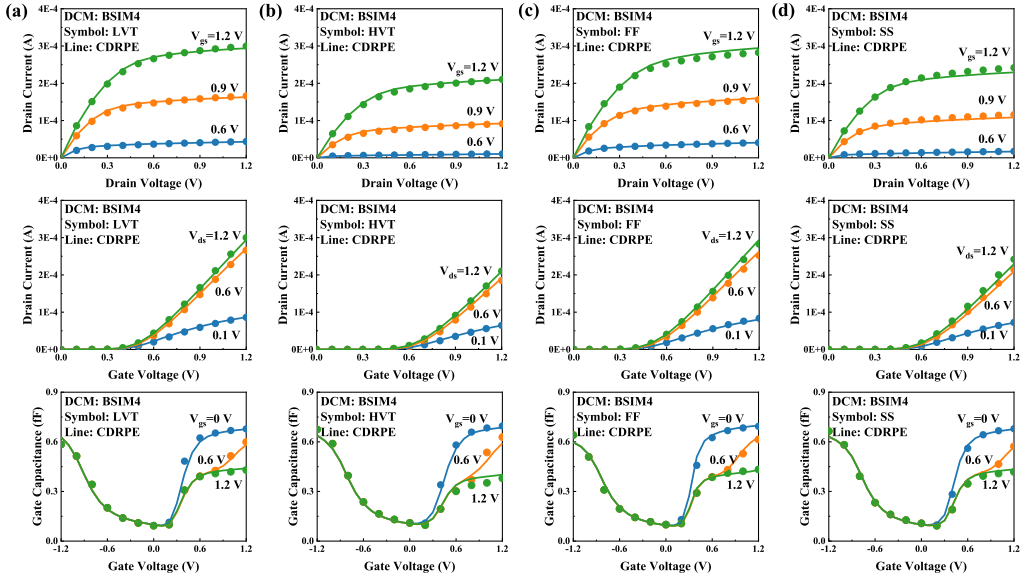


Fig. 6. Model (lines) and data (symbols) comparisons of I_d - V_d , I_d - V_g and C_{gg} - V_g for (a) LVT, (b) HVT, (c) SS, and (d) FF devices under the same technology node.

To further assess the generality of the framework across different PVT conditions within the same technology node (e.g., LVT, HVT, SS, and FF). We applied the framework to extract PDK-generated devices with $W = 0.5\mu\text{m}$ and $L = 0.1\mu\text{m}$. The extracted results are shown in Figure 6, and the average fitting RMSE is 4.2%. These results demonstrate the robustness and generalization capability of the proposed framework across diverse PVT conditions.

4.2 Model Parameter Extraction for BSIMCMG

FinFET technology has been widely adopted in advanced circuit design due to its superior advantages over traditional planar transistors. However, the complex structure of FinFET results in a significantly large number of parameters ($>1,000$) in industrial standard BSIMCMG model. To evaluate the capability of the proposed CDRPE framework in handling high-dimensional parameter spaces, a 100-parameters extraction task based on BSIMCMG is performed using TCAD simulation data [31] for 7nm technology node FinFET [32]. The core parameter set includes EOT , $PHIG$, U_0 , UA , EU , CIT , $CDSCD$, $ETA0$, $VSAT$, $KSATIV$, $PCLM$, $DVT0$, $RDSW$, $QM0$, $ETAQM$, $CGSO$, $CGSL$, $CKAPPAS$. More broadly, the extraction task involves both current-related and capacitance-related parameters. The current-related parameters consist of geometry parameters (XL , $LINT$, LL , LLN , $DELTA W$, $FECH$), accumulation charge parameters ($EOTACC$, $DELVFBACC$), mobility parameters (U_0 , UP , UA , UC , EU , UD), SCE parameters (CIT , $CDSC$, $CDSCD$, $DVT0$, $DVT1$, $\dots$), DIBL parameters ($ETA0$, $DSUB$, $PVAG$, $DROUT$, $\dots$), velocity saturation parameters ($VSAT$, $VSAT1$, $PSAT$, $PTWG$, $\dots$), channel length modulation parameters ($PCLM$, $PCLMG$), Gate-Induced-Drain-Leakage parameters ($AGIDL$, $EGIDL$, $PGIDL$, $\dots$), extension resistance parameters ($RDSW$, $RDSWMIN$), impact ionization parameters ($ALPHA0$, $BETA0$, $\dots$), and so on. The capacitance-related parameters consist of geometry parameters (DLC , $DLCACC$), quantum mechanical parameter ($QM0$, $ETAQM$, PQM , $\dots$), fringing capacitance parameters (CF , $CSDESW$), overlap capacitance parameters ($CGSO$, $CGSL$, $CKAPPAS$) and so on. The complete list of the 100 extracted parameters is provided in Appendix A.1.

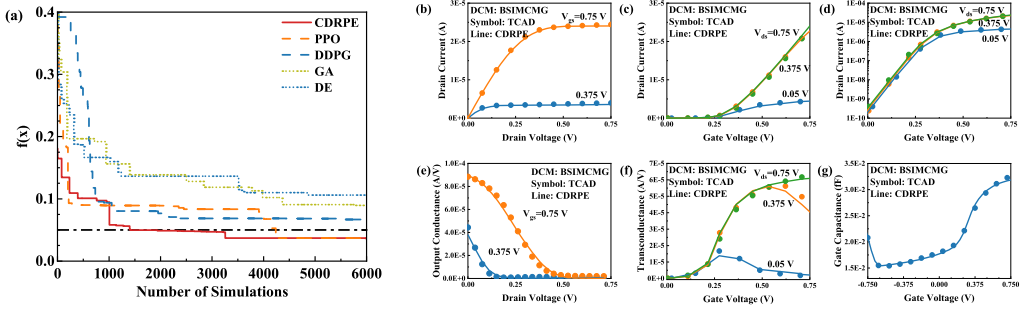


Fig. 7. (a) Performance comparisons for 100-parameter extraction task of BSIMCMG model. (b)–(g) are the model (lines) and data (symbols) comparisons of I_d-V_d , I_d-V_g , $\lg(I_d-V_g)$, $g_{ds}-V_d$, g_m-V_g and $C_{gg}-V_g$ curves.

The experimental results are summarized in Figure 7. As shown in Figure 7(a), the convergence curves demonstrate that RL-based methods exhibit superior performance with increasing parameter dimensionality compared with black-box optimization methods. In particular, DDPG, GA, and DE fail to achieve convergence within 6,000 iterations. CDRPE achieve 2.9 times faster convergence than PPO. The extracted model shows close agreement with TCAD simulated data, as illustrated in Figure 7(b)–(g), where RMSE values of $I-V$ and $C-V$ fitting is 2.82% and 4.17%, respectively. These results confirm that the proposed CDRPE framework achieves efficient and accurate parameter extraction even for large-scale tasks at advanced FinFET technology nodes.

4.3 Model Parameter Extraction for BSIMSOI

To further evaluate the generalization capability of the proposed framework in different DCMs. A 70-parameter extraction task is conducted using measured experimental data of an N-type SOI MOSFET with a width of $2\mu\text{m}$ and length of $0.1\mu\text{m}$. The core parameter set including $TOXE$, $VTH0$, $U0$, UA , CIT , $ETA0$, $VSAT$, $PCLM$, $PVAG$, $DVT0$, $RDSW$, $CGSO$, $CGSL$, $CKAPPA$. The current-related parameters consist of geometry parameters ($WINT$, $LINT$, $\dots$), mobility parameters ($U0$, UA , UB , $\dots$), SCE parameters (CIT , $CDSC$, $DVT0$, $\dots$), DIBL parameters ($ETA0$, $DSUB$, $PVAG$, $\dots$), velocity saturation parameters ($VSAT$), channel length modulation parameter ($PCLM$, $DROUT$, $\dots$), extension resistance parameters ($RDSW$), impact ionization parameters ($ALPHA0$, $BETA0$, $\dots$), self-heat parameters ($RTH0$, $CTH0$) and so on. The capacitance-related parameters consist of geometry parameters (DLC , DWC , CLC , CLE), quantum mechanical parameter ($ADOS$, $BDOS$, $MOIN$, $ACDE$), fringing capacitance parameters (CF , $CFRCOEFF$, $CSDESW$), overlap capacitance parameters ($CGSO$, $CGSL$, $CKAPPA$). The complete list of the 70 extracted parameters is provided in Appendix A.2.

The experimental results are summarized in Figure 8. As shown in Figure 8(a), the convergence curves indicate that the proposed CDRPE framework consistently outperforms traditional black-box optimization techniques and standard reinforcement learning algorithms. Figure 8(b)–(g) compare the extracted model outputs with experimental silicon data. Due to the intrinsic susceptibility of measured silicon data to electrical noise, the resulting RMSE values are slightly higher than those obtained with simulated data. The RMSE of the on-state current remains as low as 4.90%. These results demonstrate the robustness of the proposed framework in handling experimental data and highlight its practicality for real-world device modeling tasks.

To explicitly validate the extraction accuracy of secondary physical effects, specifically **Gate-Induced Drain Leakage (GIDL)**, impact ionization (II), and self-heating effect, additional experiments are performed on SOI devices. As illustrated in Figure 9(a), the inclusion of GIDL and II

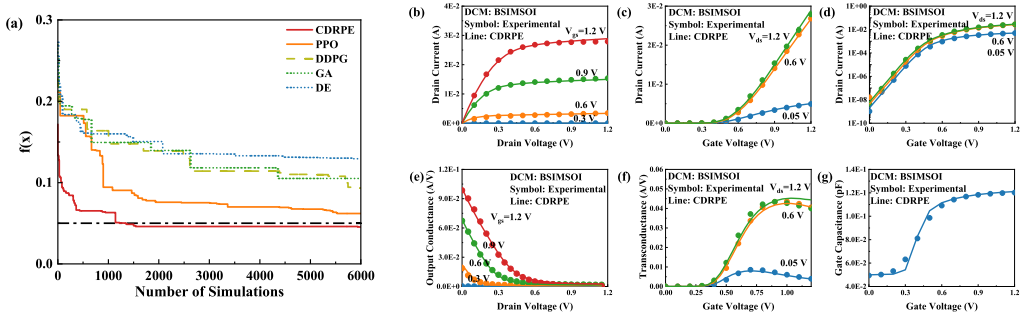


Fig. 8. (a) Performance comparisons for 70-parameter extraction task of BSIMSOI model. (b)–(g) are the model (lines) and data (symbols) comparisons of I_d – V_d , I_d – V_g , $\lg(I_d)$ – V_g , g_{ds} – V_d , g_m – V_g and C_{gg} – V_g curves.

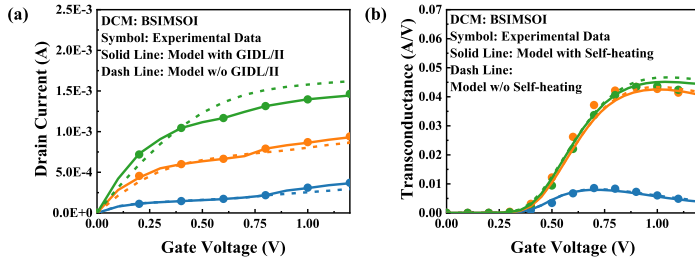


Fig. 9. Validation of secondary physical effects extraction on SOI devices. (a) Model (lines/dotted lines) and data (symbols) comparisons of output characteristics with and without GIDL and II. (b)–(g) Model (lines/dotted lines) and data (symbols) comparisons of transconductance with and without SH.

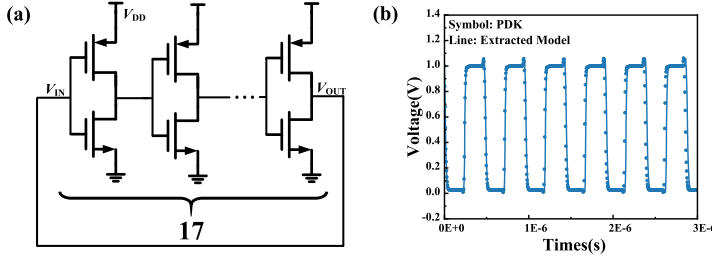


Fig. 10. (a) Schematic diagram and (b) Simulation results of RO circuits.

parameters enables the model to capture the kink effect in the saturation region of a floating-body SOI devices. Figure 9(b) compares the fitting results with and without self-heating parameters. The model extracted by CDRPE accurately reproduces these secondary physical effects, demonstrating the effectiveness of the proposed framework.

4.4 Circuit Simulation using Automatically Extracted Model

Moreover, the full cycle from model creation to circuit simulation is accomplished for both digital and analog circuits, with the results of the extracted models compared against the standard PDK models. The created BSIM4 model is employed for a 17-stage **Ring Oscillator (RO)** circuit SPICE simulation. The corresponding circuit diagram and simulation results are presented in Figure 10. The simulation process finishes within 0.05s using HSPICE without any convergence issue. The

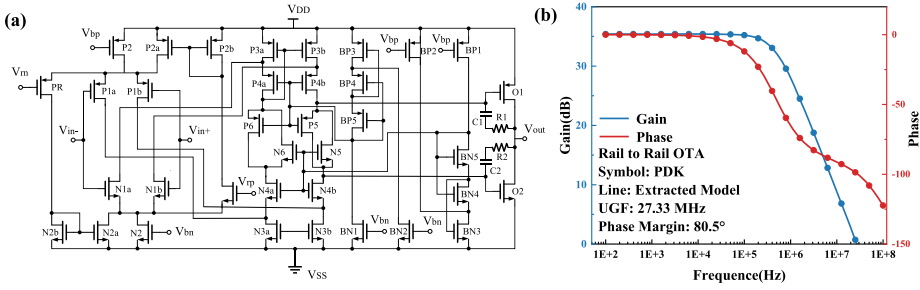


Fig. 11. (a) Schematic diagram and (b) Simulation results of rail-to-rail OTA circuits.

Table 6. Convergence Comparison of CDRPE, PPO, and Ablation Setting on the BSIMSOI Parameter Extraction Task

	CDRPE	Ablation Setting (i)	Ablation Setting (ii)	Ablation Setting (iii)	PPO
Exp-1	1,467 (2.3%)	4,587 (3.7%)	7,943 (3.6%)	2,778 (2.9%)	9,369 (4.2%)
Exp-2	1,126 (1.5%)	2,402 (2.5%)	4,681 (4.1%)	1,562 (4.1%)	16,638 (4.6%)
Exp-3	1,946 (4.2%)	6,365 (4.9%)	2,751 (4.3%)	2,726 (3.2%)	N/A (8.6%)
Exp-4	3,174 (3.5%)	1,323 (3.4%)	11,492 (4.6%)	5,310 (3.6%)	N/A (10.2%)
Exp-5	932 (3.4%)	2,164 (4.4%)	N/A (5.9%)	1,977 (4.8%)	N/A (7.3%)

extracted BSIMSOI model is used in the simulation of a rail-to-rail **operational transconductance amplifier (OTA)** with 36 MOS transistors. The schematic of OTA includes a folded-cascode and a push-pull output as shown in Figure 11(a). Figure 11(b) presents the gain bandwidth, phase margin, and other metrics obtained from this OTA circuit simulation. These results demonstrate that the automatically extracted DCMs can be seamlessly integrated into practical circuit design flows, further validating the effectiveness and reliability of the proposed extraction framework for real-world applications.

4.5 Ablation Study

To further evaluate the contribution of individual components in the proposed CDRPE framework, we conducted a series of ablation experiments by selectively removing key modules, namely (i) DL-based initialization of core DCM parameters, (ii) the self-attention network in RL agent, and (iii) device physics knowledge embedding. As summarized in Table 6, the results report the number of simulation steps required to achieve convergence ($f(x) < 0.05$) and the final RMSE achieved for CDRPE, PPO, and the three ablation settings on the BSIMSOI parameter extraction task. From the results, it can be observed that the full CDRPE framework converges significantly faster than PPO, while removing any of the three modules leads to slower convergence and higher iteration counts, confirming the effectiveness and necessity of each component in accelerating optimization. Notably, (iii) generally outperforms (i) and (ii). This indicates that the DL-based initialization and self-attention mechanism play primary roles in accelerating convergence, whereas the device physics knowledge functions as an auxiliary constraint that further enhances optimization efficiency.

Furthermore, to validate the effectiveness of the dynamic action range strategy, a sensitivity analysis is performed on the decay factor γ . As illustrated in Figure 12, a large γ maintains an overly broad action space, leading to ineffective fine-grained search near the optimum, whereas a small γ prematurely restricts exploration and increases the risk of convergence to local optima. A moderate decay factor balances these effects and ensures both rapid convergence and high precision.

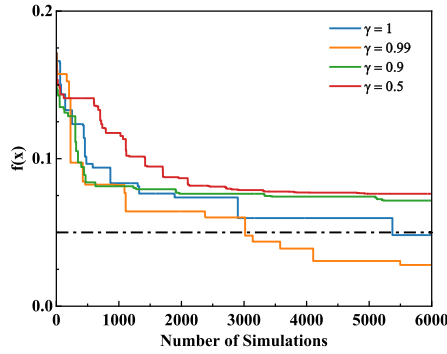


Fig. 12. Sensitivity analysis of the dynamic action range decay factor γ .

5 Conclusion

For the first time, we introduce a combined deep learning and reinforcement learning framework with continuous action space for DCM parameter extraction automation. By integrating physics-informed initialization, a self-attention-enhanced RL agent, and device knowledge embedding, the proposed method achieves fast convergence, robust stability, and physically consistent solutions. Experiments on BSIM4, BSIMCMG, and BSIMSOI demonstrate that up to 100 parameters can be extracted with less than 5% RMSE, significantly outperforming existing RL and black-box optimization methods. Furthermore, circuit-level simulations on both digital and analog benchmarks confirm the practical applicability of the created models. This work establishes an effective and scalable pathway for applying reinforcement learning to compact model development at advanced technology nodes.

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A Appendix

A.1 Extracted Parameters of BSIMCMG

Parameter Category	Extracted Parameters
Core parameters	<i>EOT, PHIG, U0, UA, EU, CIT, CDSC, ETA0, VSAT, KSATIV, PCLM, DVT0, RDSW, QM0, ETAQM, CGSO, CGSL, CKAPPAS</i>
Geometry parameters	<i>XL, LINT, LL, LLN, DELTAW, FECH, DLC, LLC</i>
Accumulation charge parameters	<i>DLCACC, EOTACC, DELVFBACC</i>
Mobility parameters	<i>UP, LPA, UC, UD, UCS, UDS, UDD, UTE</i>
Short channel effect parameters	<i>CDSCD, DVT1, DVT1SS, DVTSHIFT</i>
Drain-induced barrier lowering parameters	<i>DSUB, PVAG, DROUT, PDIBL1, PDIBL1R, PDIBL2</i>
Velocity saturation parameters	<i>VSAT1, VSAT1R, DELTAVSAT, PSAT, PTWG, PTWGR, MEXP, MEXPR</i>
Non-saturation effect parameters	<i>A1, A2</i>
Junction current parameters	<i>JSS, JSD, NJS, NJD, VFBS, VFBSDCV</i>
DITS parameters	<i>DVTP0, DVTP1</i>
Body effect parameters	<i>PHIBE, K0, K0SI, K1</i>
Self-heat effect parameters	<i>RYH0, CTH0</i>
Channel length modulation parameters	<i>PCLMG, PCLMCV</i>
Gate-induced drain leakage parameters	<i>AGIDL, BGIDL, CGIDL, EGIDL, PGIDL, AGISL, BGISL, CGISL, EGISL, PGISL</i>
Impact ionization parameters	<i>ALPHA0, ALPHA1, BETA0</i>
Extension resistance parameters	<i>RDSWMIN</i>
Quantum mechanical effect parameter	<i>PQM, QM0ACC, PQMACC, QMFACTOR, QMTCENCV</i>
Fringing capacitance parameters	<i>CFS, CFD, CSDESW</i>
Overlap capacitance parameters	<i>CGDO, CGDL, CKAPPAD, CGBO, CGBN</i>

A.2 Extracted Parameters of BSIMSOI

Parameter Category	Extracted Parameters
Core parameters	<i>TOXE, VTH0, U0, UA, CIT, ETA0, VSAT, PCLM, PVAG, DVT0, RDSW, CGSO, CGSL, CKAPPA</i>
Geometry parameters	<i>WINT, LINT, DLC, DWC, CLC, CLE</i>
Mobility parameters	<i>UB, UC, UD</i>
Short channel effect parameters	<i>CDSC, DVT1, DVT2, NFACTOR</i>
Drain-induced barrier lowering parameters	<i>ETAB, DSUB, PDIBL1, PDIBL2</i>
Bulk charge effect parameters	<i>A0, B0, AGS, KETA, KETAS</i>
Non-saturation effect parameters	<i>A1, A2</i>
Drain-induced threshold shift parameters	<i>DVTP0, DVTP1, DVTP2, DVTP3, PDITS</i>
Body effect parameters	<i>K1, K2</i>
Self-heat effect parameters	<i>RYH0, CTH0</i>
Gate-induced drain leakage parameters	<i>AGIDL, BGIDL, CGIDL, EGIDL, AGISL, BGISL, CGISL, EGISL</i>
Impact ionization parameters	<i>DELTA, ALPHA0, BETA0, BETA1, BETA2</i>
Capacitance-related parameters	<i>DELVT, MOIN, ACDE</i>
Quantum mechanical effect parameter	<i>ADOS, BDOS</i>
Fringing capacitance parameters	<i>CFRCOEFF, CSDESW</i>
Overlap capacitance parameters	<i>CGDO, CGEO, CGDL</i>

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